HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

7 December 2007

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

Only approved calculators may be used.

1. Fill in the missing values in the following table:

i	v	d	$i^{(12)}$	$d^{(4)}$
0.05	?	?	?	?

[4 marks]

2. You have a liability to make a payment of £100,000 in 20 years' time. You plan to meet this liability by investing two equal amounts of £X in a bank account now and in 10 years' time. The bank pays interest of 5% per annum effective on accounts containing up to £75,000, and 6% per annum effective on accounts containing £75,000 or more. What is X?

[6 marks]

3. (a) Show that:

   (i) $d = iv$

   (ii) $a^{i}_{\overline{n}|} = \dfrac{1 - v^n}{i}$.

[5 marks]

   (b) Evaluate the following:

   (i) $a^{0.08}_{\overline{15}|}$.

   (ii) ${}_5|a^{0.115}_{\overline{\infty}|}$.

   (iii) $(Ia)^{0.08}_{\overline{15}|}$.

[6 marks]

   (c) Find the present value, at 10% per annum interest effective, of an immediate annuity, payable annually in arrear for 20 years. The first payment is £1,000, and subsequent payments *decrease* by 3% compound each year [meaning that each payment is less than the previous payment by a factor 1/(1.03)].

[4 marks]

[Total 15 marks]

continued ...

4. (a) You borrow £100,000 to buy a flat. The loan is to be repaid over 25 years by level annual repayments, payable in arrear. The interest rate is 7% per annum effective.

   (i) What is the level annual repayment?

   (ii) What is the capital outstanding just after the 9^{th} repayment has been made?

   (iii) What is the interest content of the 10^{th} repayment?

   [5 marks]

   (b) Just after the 10^{th} repayment has been made, you run into financial difficulties, such that you cannot afford to continue to make repayments of more than £7,500 per year. You ask the lender if you can extend the term of the loan so that your annual repayment does not exceed this amount. What is the shortest possible extension of the term that will meet your needs?

   [5 marks]

   [Total 10 marks]

5. (a) State the Principle of Equivalence, and explain how it may be used as a criterion for deciding whether or not to buy an investment.

   [2 marks]

   (b) You are deciding which of three investments to buy, A, B or C. All have the same price, £10,000.

   - Investment A will pay a lump sum of £26,000 at the end of 10 years.
   - Investment B will pay £110 at the end of each of the next 10 years, plus a lump sum of £10,000 at the end of the 10 years.
   - Investment C will pay £1,800 at the end of each of the next 10 years.

   Find the yield on each investment, and on that basis, state which of A, B or C would you prefer.

   [8 marks]

   [Total 10 marks]

continued ...

6. A corporation issues a bond with term 15 years, to be redeemed at a discount of 3%, with a coupon of 8% per annum payable half-yearly in arrear.

   (a) Some of the bond is purchased at outset by investor A who pays income tax at 20% and capital gains tax at 40%. Investor A pays a price P per £100 nominal, such that their net redemption yield is 4% per annum effective. What is P?

   [5 marks]

   (b) Immediately after payment of the 20^{th} coupon, investor B offers to buy the bond from investor A for £P (that is, the same price for which investor A originally bought the bond). State with reasons whether investor A should accept the offer, or instead should hold the bond to redemption.

   [3 marks]

   [Total 8 marks]

7. An investment fund has assets worth £2,000,000 on 31 December 2006. It receives an inflow of £150,000 on 1 January 2007 and pays out £350,000 on 1 January 2008. On 31 December 2008 it has assets of £1,700,000. Then on 1 January 2009 it pays out a further £100,000. On 1 July 2010 it has assets worth £2,000,000.

   (a) Find the linked internal rate of return between 31 December 2006 and 1 July 2010, based on the subdivision of the period into 2007–2008 and 2009–2010.

   [5 marks]

   (b) Suppose also that the fund has assets of £2,250,000 on 31 December 2007. Find the time weighted rate of return during the period 31 December 2006 to 1 July 2010.

   [2 marks]

   [Total 7 marks]

Total 60 marks